AnchorMTM: Spatially Anchored Merge Tree Maps with Position Bounds

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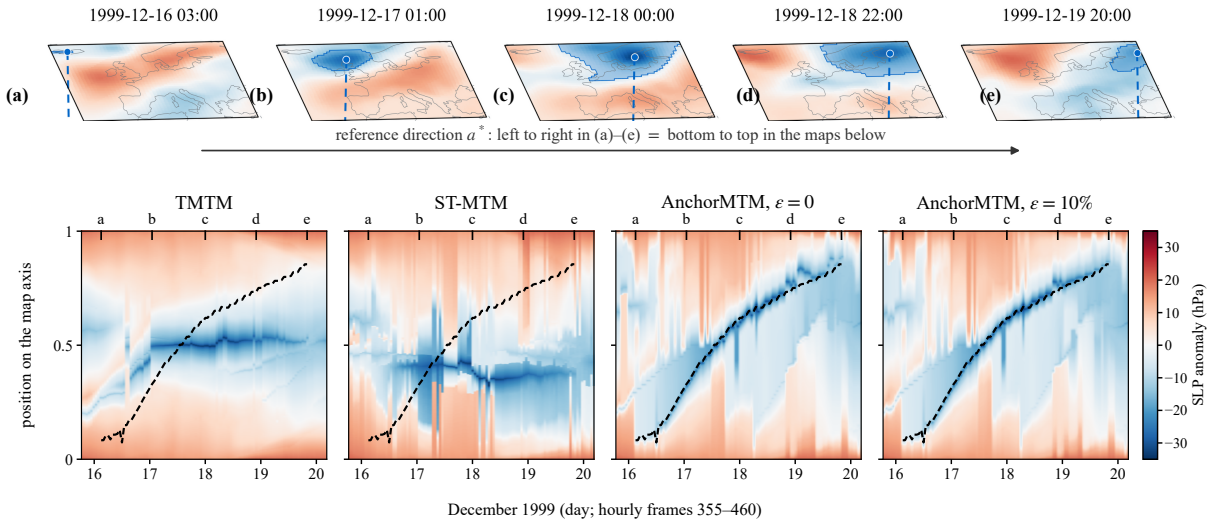


Figure 1: One storm in four merge tree maps. (a)–(e) Sea-level-pressure anomaly over the North Atlantic and Europe, 16–19 December 1999. The marked low travels from the open Atlantic to Scandinavia, that is, from left to right along the reference direction a^*. Below, four maps draw the same 106 hourly frames; the dashed line is the storm's projected location, where a map whose axis encodes location should draw it. TMTM and ST-MTM keep the storm near the middle of the axis. AnchorMTM follows it with the exact tree ($\varepsilon = 0$; mean deviation 1.5% of the axis) and with a merge-error budget of 10% (1.0%).

Abstract

Merge tree maps summarize a time-varying scalar field in one static image. Each time step becomes a one-dimensional profile of its merge tree, and the profiles are stacked along time, so that nested features appear as nested bands. The vertical axis of these maps, however, has no spatial meaning: an analyst sees how features nest, but not where they are or where they move. We present AnchorMTM, a merge tree map whose vertical axis is a fixed spatial direction, along which every extremum is drawn near the projection of its location. Keeping the tree on such an axis has a price. An exact profile keeps every subtree in one contiguous block, whereas the projected locations of sibling subtrees may interleave. We derive three lower bounds on the displacement this conflict forces, for an exact tree, for features of finite width and for a bounded change of merge levels. The bounds depend only on the data, so they are known before any map is drawn. AnchorMTM computes the map directly from the field sequence and one parameter, a merge-error budget that limits how much merge-level fidelity may be traded for position. It attains the bounds on 606 of 614 benchmark frames. The bounds also show that 31–95% of the order error of existing merge tree maps is not forced by the tree. With the tree kept exactly, AnchorMTM lowers the mean worst displacement of TMTM and ST-MTM by 73–96% on five planar sequences and by 63–67% on two volumes, while ST-MTM keeps better neighborhood preservation.

Index Terms: Time-varying scalar fields, merge tree maps, topology preservation, spatial fidelity, lower bounds, constrained layout.

1 Introduction

Time-varying scalar fields record how quantities such as sea-level pressure, aerosol concentration or vorticity evolve over a spatial domain. Analysts study them through features, such as low-pressure systems, pollution episodes or vortices, and ask three questions about each feature: how it relates to the others, where it is, and how it moves. Animation keeps every time step in its spatial context but makes distant times hard to compare [47, 53]. Static overviews therefore give one display axis to time and compress space into the other, so that a whole sequence fits into one image [2]. The Hovmöller diagram of atmospheric science is the classic example: its second axis is a physical coordinate such as longitude, and an eastward-moving system appears as a slanted streak [26]. To obtain this axis, the diagram averages the field over the remaining coordinates, and features at the same longitude blend into one.

Merge tree maps answer the first question, which the Hovmöller diagram cannot. A merge tree records how the sublevel sets of a field connect as the threshold rises: its leaves are the extrema, and its internal nodes are the levels at which regions merge [22]. Temporal merge tree maps (TMTM) draw the tree of each time step as a one-dimensional profile, whose minima are the extrema and whose barriers are the merge levels, and stack the profiles into one image [31]. Spatiotemporal merge tree maps (ST-MTM) additionally

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arrange the features by their pairwise distances [10]. The second question remains open. The vertical axis of these maps follows the tree and temporal coherence, or relative distances, and neither fixes an origin or a direction in space. Figure 1 shows the consequence. A storm crosses the North Atlantic within three days, yet both maps draw it near the middle of the axis throughout; the reader sees how the storm connects to neighboring lows, but not where it is or which way it moves.

We give the vertical axis a spatial meaning. We fix a direction in the domain and assign every extremum a reference position, the projection of its location onto this direction; a map that encodes location should draw each extremum at its reference. The obstacle is the tree itself. A profile shows the tree exactly only if every subtree occupies one contiguous block, but the references of two sibling subtrees may interleave (Fig. 2). In that case no exact profile can place every feature at its reference, whatever algorithm draws it. The conflict raises three questions: how much displacement the tree forces, where in the tree this displacement originates, and how much merge-level fidelity must be given up to remove it. Without answers, the positional error of a map cannot be attributed to the data or to its layout, and an analyst cannot judge whether a spatial axis is worth its cost.

AnchorMTM answers these questions and draws the map. It takes the field sequence and one number, the merge-error budget ε, and computes the map directly; it does not post-process the layout of another map. The reference direction is estimated from the motion of the extrema or chosen by the analyst. From the trees and the references alone, three lower bounds give the displacement that an exact tree, finite feature widths and a bounded merge error force, so the price of the spatial axis is known before any layout is computed. AnchorMTM then chooses, frame by frame, the leaf order with the least attainable displacement among the orders that fit the budget, places the extrema as close to their references as this order allows, and fills the profile. With $\varepsilon = 0$, the map shows every tree exactly; with $\varepsilon > 0$, it moves features closer to their references and reports the merge error that each frame actually spends.

Our contributions are as follows.

- We propose AnchorMTM, a merge tree map whose vertical axis is a fixed spatial direction. It builds the leaf order and the extremum positions of every frame from the projected locations of the extrema, so that the height of a band reads as location and its drift as motion along this direction, while the profile keeps the nesting and the merge levels of the tree.
- We formalize the conflict between keeping the tree and keeping location, and derive three lower bounds on the displacement it forces, for an exact tree, for finite feature widths and for a bounded merge error. The bounds require no layout; they locate the conflict in the tree and measure any merge tree map, including existing ones.
- We introduce a merge-error budget ε that limits how much merge-level fidelity may be traded for position. A closed-form test decides whether a leaf order fits the budget and yields its optimal merge barriers, so one parameter controls the whole construction, and every rendered frame is checked against it.
- On five planar and two volumetric sequences, the bounds are attained on 606 of 614 planar frames, most of the order error of existing maps proves avoidable, and at the same topological guarantee AnchorMTM draws features far closer to their locations.

2 Related Work

2.1 Temporal Overviews of Spatial Data

Animation preserves the spatial context of each time step, whereas static overviews let viewers compare distant times directly; which serves better depends on the task [47, 53]. Bach et al. describe both, and the operations between them, as transformations of a space–time cube [2]. A static overview must linearize space. Hovmöller diagrams keep one physical coordinate and average over the others [26]. Geographic orderings, dense pixel displays and space-filling curves traverse the whole domain [9, 21, 46], and volume lines and data-driven curves extend such traversals to volumes [59, 66]. For moving objects, one-dimensional projections and summaries such as MotionRugs, stable visual summaries and MoReVis order objects by position or by stability [6, 19, 54, 62]. Every linearization distorts some spatial relations [1, 41]. These overviews keep a spatial coordinate but no feature hierarchy; AnchorMTM keeps the merge tree and states what the spatial coordinate then costs.

2.2 Topology of Time-Varying Scalar Fields

Merge trees and contour trees summarize how level sets connect, and persistence-based simplification controls their size [14, 22, 52]. Features are tracked over time by region overlap or attribute similarity [48, 49] and shown as tracking graphs and nested tracking graphs [37, 38, 60]. Time-varying merge trees have been computed and compared with Wasserstein and geometry-aware interleaving distances [42, 44, 63, 65], and such pipelines follow tropical cyclones and atmospheric rivers [32, 64]. The merge error we bound is a labelled interleaving distance [20, 39, 40] and relates to distances between ordered merge trees [5]. Tracking graphs show how features appear, split and merge, but not where they are; a merge tree map also shows the values and the nesting of every time step, and AnchorMTM adds their location.

2.3 Drawing Hierarchies in One Dimension

Topological landscapes draw a merge tree as a terrain or as a one-dimensional profile whose minima are the extrema and whose barriers are the merge levels [43, 57]; geometry-preserving and temporally coherent variants exist [4, 24], and contour trees have also been drawn in the plane and in joint layouts [23, 36]. Temporal treemaps lay out evolving hierarchies along time [30]. TMTM aligns the profiles of successive merge trees into one static map [31], and ST-MTM adds relative spatial distances and temporal stability [10]. Storyline layouts face a related problem for groups of entities and balance crossings and wiggles against group contiguity [35, 51]. In all of these layouts, a position on the axis encodes a tree traversal, a temporal alignment or a relative distance. AnchorMTM makes it encode location.

2.4 Hierarchies against an External Order

The conflict we study arises whenever a tree must respect an external order. Optimal leaf ordering selects a tree-consistent order of a dendrogram that keeps similar leaves adjacent [3, 15], and tree reordering seeks the tree-consistent permutation closest to a target order [7]. Spatially ordered treemaps and data–spatial grid maps trade hierarchy or data layout against geographic position [55, 61], geophylogenies connect leaf orders to map locations [28], and tanglegram layouts minimize crossings or displacement between two trees [18, 27]. Separation constraints keep laid-out objects disjoint [13, 34], ultrametric fitting under the maximum norm is closely related to our optimal barriers [17], and merge distortion and topology-preserving projections measure how much hierarchy a mapping retains [12, 16]. These methods optimize crossings, sums or stress for a given instance. We instead bound the worst displacement that every tree-consistent layout must accept, in closed form for points, together with the merge error that removes it.

3 Spatially Anchored Merge Tree Maps

This section defines the map and the two errors that describe each of its frames, fixes the spatial reference, and shows where the refer-

ence conflicts with the tree.

3.1 Merge Trees and One-Dimensional Profiles

For a scalar field f_t on a 2D or 3D domain, the merge tree T_t records how sublevel-set components appear at minima and join at saddles as the threshold increases; negating the field gives the split tree [22, 52]. We simplify each tree by persistence before layout and refer to the simplified tree throughout [14]. Each leaf i is an extremum at location p_i^t. Its support S_i^t is the set of samples on its leaf arc, with measure m_i^t, and two leaves merge at the level $V_{ij}^t = h_t(\mathrm{lca}(i,j))$ of their lowest common ancestor; V_{ii}^t is the extremum value, and equal-level merges form one multifurcation.

A one-dimensional profile draws the leaves as minima in a chosen leaf order, each inside an interval sized by its support, and fills the space between them from the tree, so that the largest profile value between two minima shows their merge level [31, 43, 57]. A profile shows the tree exactly only if every subtree occupies one contiguous block of leaves. Stacking the profiles along time gives the map. Leaves of consecutive frames are matched by support overlap with a one-to-one assignment, and $\mu_t(i)$ denotes the matched predecessor of leaf i [10].

3.2 Two Kinds of Fidelity

A frame of the map consists of a leaf order π, an interval for each leaf with center z_i and width W_i, an anchor x_i at which the extremum is drawn, and the profile ψ. The intervals follow π on the canvas $[0,L]$, are separated by gaps of at least g, and hold each anchor within $\rho W_i/2$ of its center; $\mathscr{L}(\pi;W)$ denotes these placements. A frame can be wrong in two ways. First, its profile can misstate merge levels. The *merge error* compares displayed and true levels for fixed labels,

$$d_{\mathrm{top}}(\psi,T) = \max_{i,j} \left| V_{ij}^{\psi} - V_{ij} \right|, \tag{1}$$

where V_{ij}^{ψ} is the largest profile value between the anchors of i and j. It is zero exactly when the profile shows the tree, and for complete labels it equals the labelled interleaving distance, which bounds the interleaving and bottleneck distances [20,39,40]. Second, the frame can draw an extremum away from its reference position q_i, which we define below. We measure this *displacement* by its worst value $\|x-q\|_\infty$ per frame.

How much of the displacement is due to the leaf order alone? Two quantities answer this:

$$\begin{aligned} \mathrm{NROE}(\pi) &= \frac{1}{2L} \max_{j \le k} \left[q_{\pi(j)} - q_{\pi(k)} \right]^+, \\ \tau(\pi) &= \min_{(x,z) \in \mathscr{L}(\pi;W)} \|x-q\|_\infty . \end{aligned} \tag{2}$$

The *order error* NROE is the least worst displacement of points drawn in order π, relative to L: if a leaf precedes another whose reference is lower by Δ, one of the two must move by at least $\Delta/2$. The interval cost τ is the same quantity for intervals of the given widths; it is computed exactly for a fixed order.

The two errors compete, and a spatially anchored map must decide how to balance them. We therefore state the construction of each frame as one problem: bring the anchors close to their references and keep them stable in time, subject to a declared limit η on the merge error,

$$\begin{aligned} \min_{\pi,\,(x,z) \in \mathscr{L}(\pi;W)} \; & \lambda_R E_R + \lambda_G E_G + \lambda_M E_M + \lambda_C E_C \\ \text{s.t. } & d_{\mathrm{top}}(\psi,T) \le \eta . \end{aligned} \tag{3}$$

Here E_R is the mean squared distance of anchors to their references, E_G a stress between anchor separations and support-centroid distances, E_M the mean squared distance to a temporal target $\hat{x}_i$ derived from the matched previous anchor, and E_C the offset of anchors from interval centers. The allowance η makes the trade-off explicit: $\eta = 0$ demands the exact tree, and a larger η admits leaf orders that follow the references more closely. Existing merge tree maps solve problems of this kind with $\eta = 0$ and without the reference term. TMTM orders children by their temporal overlap and packs intervals by sample count [31], and ST-MTM chooses a temporally concordant or optimal leaf order and fits relative distances [10]. Without E_R, nothing ties their axes to location. The supplement relates every compared method to Eq. 3.

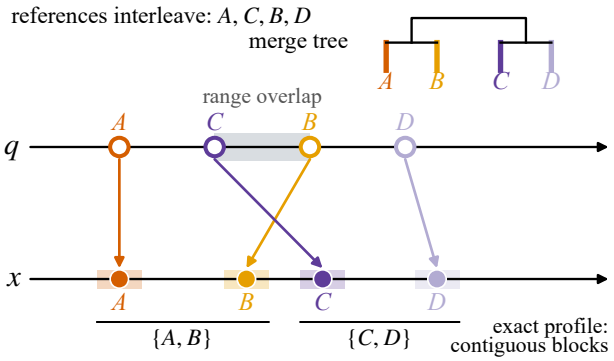


Figure 2: Reference–topology conflict. The merge tree requires the blocks $\{A,B\}$ and $\{C,D\}$, but the references occur in the order A,C,B,D, so every exact profile must move some anchors away from their references (arrows). The overlap of the two subtrees' reference ranges (grey) determines the least displacement (Theorem 1). Positions are schematic.

3.3 Spatial Reference

The reference gives the axis its meaning. For one unit direction a, the reference position of leaf i is $q_i^t = \langle a, p_i^t \rangle - o$, where the origin o and the extent L come from the projected corners of the domain. By default, a is the direction along which the matched extrema move most,

$$a^* \in \arg\max_{\|a\|=1} \; a^\top \Big(\sum_{t,i} \Delta p_i^t (\Delta p_i^t)^\top \Big) a, \tag{4}$$

where $\Delta p_i^t = p_i^t - p_{\mu_t(i)}^{t-1}$ is the motion of a matched extremum, signed so that the total projected motion is nonnegative; an analyst may instead choose the direction. Direction, origin and extent stay fixed for the whole sequence and for every budget, without per-frame recentering. A feature that moves along a therefore moves along the axis, and a feature that stays in place keeps its height.

3.4 Where Topology and Space Conflict

Figure 2 shows the smallest instance of the conflict. The tree joins A with B and C with D, so an exact profile must draw $\{A,B\}$ and $\{C,D\}$ as two contiguous blocks. The references, however, occur in the order A,C,B,D: C lies between A and B, and B between C and D. Neither order of the two blocks matches, so some anchors must leave their references, and finite widths add further packing constraints. How far they must move depends only on the tree, the references, the widths and the allowance, not on the algorithm that draws the map. The next section computes it.

4 The Positional Price of Keeping the Tree

We now answer the three questions of the introduction in turn: what an exact tree forces when features are points, what finite widths add,

and what a merge allowance releases. Each bound involves only T, q, W and η, so it holds for every solution of Eq. 3, whatever its weights or its order rule. The supplement contains the full proofs. Within a frame we drop t, write $[v]^+ = \max(v,0)$, and take empty maxima as zero.

4.1 Exact Topology

Consider two sibling subtrees X and Y, whose leaves have references in the ranges $[m_X, M_X]$ and $[m_Y, M_Y]$, on the reduced tree in which equal-level merges form one node. An exact profile must draw one block entirely before the other. If X comes first, its highest reference M_X precedes the lowest reference m_Y of Y, and by the argument behind NROE one of the two leaves moves by at least $[M_X - m_Y]^+/2$; the other orientation costs $[M_Y - m_X]^+/2$. Even the cheaper orientation pays half the overlap that remains, and the worst sibling pair in the tree sets the price.

Theorem 1 (Exact-Topology Position Bound)**.** *Every profile with $d_{\text{top}} = 0$ has $\text{NROE}(\pi) \geq \nu^*(T,q)$, where*

$$\nu^*(T,q) = \frac{1}{2L} \max_{v} \max_{\substack{X,Y \in \text{ch}(v) \\ X \neq Y}} \big[\min(M_X - m_Y, M_Y - m_X)\big]^+. \quad (5)$$

Ordering the children of every node by the midpoints of their reference ranges attains ν^* in $O(n \log n)$ time.

Proof sketch. Midpoint order chooses the cheaper orientation of every sibling pair, and every inverted pair of leaves is separated by two sibling blocks at its lowest common ancestor, so no inversion costs more than the pair that defines ν^*. □

Theorem 1 answers the first two questions for points. The price of an exact tree is half the reference overlap of the worst pair of siblings, and the maximizing node shows where in the tree this price arises. In Fig. 2, the ranges of $\{A,B\}$ and $\{C,D\}$ overlap by $q_B - q_C$, so every exact profile has an order error of at least $(q_B - q_C)/2L$.

Because the midpoint order of the same tree attains ν^*, the bound also splits the order error of any exact-topology map into two parts: ν^*, which the tree forces, and $\text{NROE}(\pi) - \nu^*$, which the order rule of the map chooses and which another order of the same tree would remove. NROE ignores widths, so this split applies equally to maps whose intervals follow different rules.

4.2 Finite Widths

Features are not points. A block of leaves occupies at least the sum of its widths and gaps, so even two sibling blocks in their better orientation may have to push anchors away from their references, and the canvas ends add the same kind of pressure. For two disjoint leaf sets X and Y, let $\omega(X,Y)$ be the least worst displacement that drawing all of X before all of Y, or the reverse, forces on their anchors, and let E_∂ be the least displacement needed to fit all intervals between the canvas ends. Both follow from sorted reference endpoints and cumulative widths.

Theorem 2 (Width-Aware Position Bound)**.** *Let $\mathscr{P}_0$ contain every pair of descendant-leaf sets of two siblings. Every exact-topology interval layout satisfies*

$$\begin{aligned} \|x - q\|_\infty \geq \tau(\pi) &\geq \mathscr{B}_w(T,q,W) \\ &= \max\Big\{E_\partial, \max_{(X,Y)\in\mathscr{P}_0} \omega(X,Y)\Big\}. \end{aligned} \quad (6)$$

With zero widths and gaps, $\mathscr{B}_w = L\,\nu^*(T,q)$.

Proof sketch. The first and last anchors of a block must enclose its intervals and gaps up to the anchor slack, which leaves a shortfall against their reference separation; the exact tree fixes one orientation for every sibling pair, and the canvas ends give one-sided costs. □

Unlike ν^*, the width bound $\mathscr{B}_w$ need not be attained. When a map's interval cost exceeds it, the excess may lie partly in the bound itself.

4.3 A Bounded Merge Error

Suppose the profile may misstate merge levels by up to η. Does this let it interleave subtrees freely? It does not. If a leaf k is drawn between two descendants i and j of a node u, the profile between i and j passes k, so the displayed merge of i and k cannot exceed that of i and j; with errors of at most η on both, $V_{ik} \leq h(u) + 2\eta$. Leaf k may therefore enter the block of u only if it descends from $A_\eta(u)$, the highest ancestor of u whose level lies within 2η of $h(u)$. Let $\mathscr{P}_\eta$ contain the pairs of leaf sets that this rule keeps apart: $(\mathscr{D}(u), \{k\})$ for leaves k outside $\mathscr{D}(A_\eta(u))$, and $(\mathscr{D}(u), \mathscr{D}(w))$ for internal nodes u, w with $\mathscr{D}(w)$ disjoint from $\mathscr{D}(A_\eta(u))$ and $\mathscr{D}(u)$ disjoint from $\mathscr{D}(A_\eta(w))$, where $\mathscr{D}(\cdot)$ denotes descendant leaves.

Theorem 3 (Bounded-Topology Position Bound)**.** *In every profile with $d_{\text{top}} \leq \eta$, each pair in $\mathscr{P}_\eta$ is drawn wholly in one orientation. Hence*

$$\text{NROE}(\pi) \geq C_\eta = \max_{(X,Y)\in\mathscr{P}_\eta} \frac{[\min(M_X - m_Y, M_Y - m_X)]^+}{2L}, \quad (7)$$

and $\tau(\pi)$ is bounded by the expression of Theorem 2 with $\mathscr{P}_\eta$ in place of $\mathscr{P}_0$.

Theorem 3 answers the third question. A subtree may interleave with its sibling once 2η reaches the gap between its own merge level and that of its parent. Conflicts between subtrees that merge just below their parent are therefore cheap to release, whereas conflicts at deep merges stay expensive. Flattening every parent gap of at most 2η gives a second valid bound; neither bound dominates the other, and we report the larger. Read in reverse, the theorem also prices the opposite extreme: a map whose order error is below C_η, such as a projection with zero order error, must show a merge error above η.

5 AnchorMTM

AnchorMTM solves Eq. 3 for every frame of a sequence, with the reference term and the allowance $\eta = (\varepsilon_0 + \varepsilon)\Delta_F$. Here Δ_F is the scalar range of the sequence, $\varepsilon_0 = 10^{-7}$ a fixed numerical tolerance, and $\varepsilon \geq 0$ the budget, its only control. *Exact Mode* ($\varepsilon = 0$) keeps the tree up to ties within $\varepsilon_0\Delta_F$, and *Budgeted Mode* ($\varepsilon > 0$) trades a declared merge error for position. Figure 3 shows the three stages. The first stage computes the trees, the matches and the reference as described above, and gives every leaf a width $W_i = s\,m_i$ with one fixed scale s. Eq. 3 couples a discrete leaf order with continuous placement, and the second stage separates them in the way the bounds suggest: it chooses the order by τ, the quantity that the bounds constrain, and only then places the anchors. The third stage draws and checks the profiles.

5.1 Leaf Order within the Budget

Which leaf orders may a frame use? An order is admissible if some choice of the profile between its anchors keeps the merge error within η. This can be decided exactly, because for a fixed order the best barriers have a closed form.

Proposition 1 (Optimal Barriers)**.** *For a fixed order, let $C_k = \min_{i \leq k < j} V_{\pi(i)\pi(j)}$ be the lowest merge across gap k and F_k the larger of its two adjacent extremum values. The smallest attainable merge error is*

$$\begin{aligned} \delta^*(\pi) = \max\Big\{0,\ &\max_k (F_k - C_k), \\ &\tfrac{1}{2}\max_{i<j}\big(V_{\pi(i)\pi(j)} - \max_{i \leq k < j} C_k\big)\Big\}, \end{aligned} \quad (8)$$

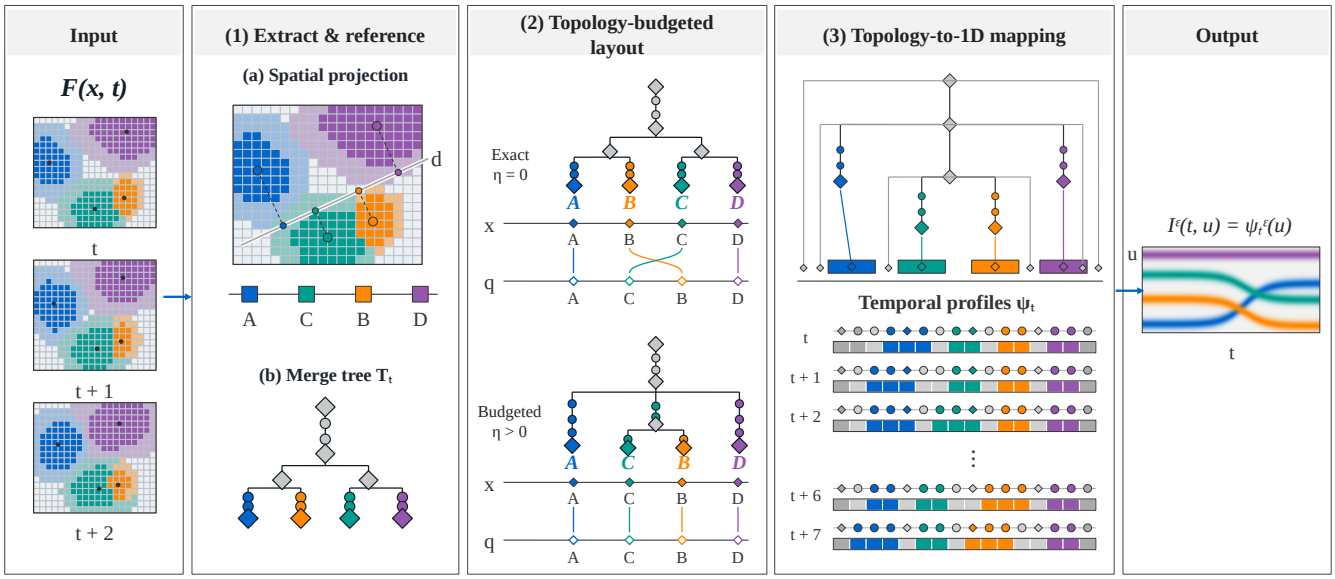


Figure 3: AnchorMTM construction ($F(x,t)$: the field at location x and time t). (1) Extract and reference: merge trees of each frame, leaf supports matched across frames, and one fixed direction d (the a^* of Eq. 4) that gives every extremum its reference position q. (2) Topology-budgeted layout: leaf orders admitted by the allowance η, then anchors x placed under a displacement cap; with $\eta = 0$ the order keeps the subtree blocks and anchors leave their references, whereas $\eta > 0$ admits an order that follows them. (3) Topology-to-1D mapping: intervals and barriers fill each profile ψ_t, and the profiles form the map $I^\varepsilon(t,u) = \psi_t^\varepsilon(u)$. Diagrams are schematic.

attained by the barriers $b_k = C_k + \delta^*(\pi)$.

Proposition 1 turns the budget into a test, $\delta^*(\pi) \le \eta$, and fixes the barriers once an order is chosen. This is what lets a single number control the construction. AnchorMTM evaluates a finite pool of candidate orders against the original merge matrix: the exact tree in midpoint order, which attains ν^*; flattenings of the tree at thresholds and at conflicting merges; and the orders visited while width-aware block moves refine the best of them. Among the admissible candidates it selects the one with the least interval cost τ and breaks ties by NROE. The pool is heuristic, so the selected order is not certified optimal over all permutations; the evaluation measures how often it attains the bounds.

5.2 Anchor Placement

With the order fixed, Eq. 3 becomes a convex quadratic program. AnchorMTM uses the weights $(\lambda_R, \lambda_G, \lambda_M, \lambda_C) = (4, 1, 0.5, 0.1)$ and the temporal target $\hat{x}_i^t = x_{\mu_t(i)}^{t-1} + (q_i^t - q_{\mu_t(i)}^{t-1})$, which moves an anchor with its projected motion and keeps its displacement $x - q$ continuous over time; leaves without a matched predecessor have no temporal term. The program also obeys the cap $\|x - q\|_\infty \le \tau(\pi) + \beta_0$ with a fixed slack β_0. The cap is feasible because τ is attained, and it passes the positional guarantee of the order on to the anchors: the other terms shape the layout but cannot undo what the order achieved.

5.3 Profile Reconstruction and Checks

The profiles reuse the leaf-arc sampling and the filling along paths to the lowest common ancestor of earlier merge tree maps [10, 31]. Extrema stay at their anchors, and each gap is then clipped or raised to its barrier b_k. All frames share the direction, origin, extent and pixel scale, so positions are comparable across time and across budgets. Raster rounding and strict-extremum clearance can still change a displayed merge level, so every rendered frame is checked against the original merge matrix, and each frame records the requested budget, τ, δ^* and the measured errors. Capacity, input and solver failures are reported rather than repaired by rescaling.

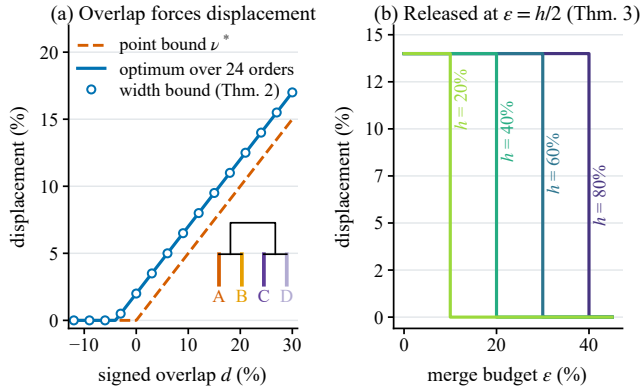


Figure 4: Controlled four-leaf trees in the configuration of Fig. 2 (inset). (a) Increasing the overlap d of the sibling ranges forces displacement; the width bound of Theorem 2 (circles) equals the optimum over all 24 leaf orders. (b) At $d = 24\%$, the 14% displacement disappears exactly when the merge budget reaches half the parent gap h, for all four gaps, as Theorem 3 predicts. Percentages refer to a unit canvas or a unit scalar range.

Computation. AnchorMTM needs no prior layout: given the fields and ε, every step above is deterministic. The bounds take at most 5.5 ms per frame on an Apple M2 Pro. One complete run on a benchmark sequence takes 0.5–63 s on an Apple M4, of which ordering and placement take 0.15–3.3 s; the rest is tree extraction, matching, rasterization and checks. The 744-frame hourly $PM_{2.5}$ sequence takes 231 s at $\varepsilon = 5\%$.

Table 1: Five-dataset comparison. Bold/underline mark the best/second distinct local measurements per dataset; ties share emphasis. Curves use fixed default orientations.

Dataset	Hilbert			Morton			TMTM			ST-MTM			AnchorMTM (ours)			
	SNS↓	TW↑	NPE↓	SNS↓	TW↑	NPE↓	SNS↓	TW↑	NPE↓	SNS↓	TW↑	NPE↓	Mode	SNS↓	TW↑	NPE↓
Gaussian	0.1117	0.6719	0.7557	<u>0.0393</u>	<u>0.8333</u>	0.7359	0.2500†	0.7700†	<u>0.1118</u>	0.0600†	1.0000†	0.1154	Exact	**0.0353**	**0.9323**	**0.0089**
													Balanced	**0.0353**	**0.9323**	**0.0089**
Ring	**0.1252**	**0.9048**	0.5941	0.1489	0.7619	0.6033	0.2800†	0.6800†	0.1756	0.2100†	0.8800†	0.2463	Exact	0.1485	<u>0.8571</u>	<u>0.0471</u>
													Balanced	<u>0.1334</u>	<u>0.8571</u>	**0.0395**
Wildfires	0.1883	0.8303	0.3641	0.1304	0.8758	0.1560	0.4400†	0.8800†	0.2280	0.2400†	0.9100†	0.3180	Exact	**0.0426**	**0.9136**	<u>0.0131</u>
													Balanced	<u>0.0447</u>	<u>0.8989</u>	**0.0116**
Storms	0.1290	0.7403	0.3705	0.1180	0.7251	0.3468	0.3400†	0.7100†	0.5753	0.2100†	0.9400†	0.4579	Exact	<u>0.0790</u>	<u>0.8463</u>	0.0399
													Balanced	**0.0743**	**0.8766**	**0.0140**
$PM_{2.5}$	0.2386	**0.8887**	0.8090	<u>0.1760</u>	<u>0.8277</u>	0.8823	0.3018	0.7781	0.5241	0.1897	0.8224	0.5577	Exact	0.1964	0.7884	<u>0.1390</u>
													5%	**0.1636**	0.7657	**0.0468**

†Grey: published TMTM/ST-MTM Temporal SNS/TW on the four benchmark inputs (ST-MTM Table 1, authors' inputs), shown for reference and excluded from emphasis; all other entries, including every $PM_{2.5}$ value, are local measurements on the same inputs. Author inputs and coverage are unconfirmed. Local TW coverage (Gaussian/Ring/Wildfires/Storms/$PM_{2.5}$) is 64/64, 1/40, 4/32, 11/744, 744/744; ineligible frames are omitted. Published TW has two-decimal precision. Mode applies only to AnchorMTM; $PM_{2.5}$ uses a fixed 5% budget, not a selected Balanced budget.

Table 2: Two groups of sequences. The bound suite has saved trees and serves the bounds; the layout suite recomputes the trees and serves the comparison of complete maps.

Suite	Sequence	Field	Frames
Bound	Storms [25]	ERA5 sea-level pressure (12 h)	118
	Wildfires [19]	fire radiative power, daily	30
	OLR [33]	outgoing longwave radiation	60
	$PM_{2.5}$ [11]	$PM_{2.5}$ concentration, daily	366
	Ring	synthetic ring	40
	Tangaroa [45]	velocity magnitude (3D)	201
	Cylinder [8]	Q-criterion (3D)	102
Layout	Gaussian	three moving peaks	64
	Ring	synthetic ring	40
	Wildfires	fire radiative power (32 bins)	32
	Storms	ERA5 pressure anomaly (1 h)	744
	$PM_{2.5}$ [29]	$PM_{2.5}$ concentration, hourly	744
	Tangaroa	velocity magnitude (3D)	201
	Cylinder	Q-criterion (3D)	102

6 Evaluation

We test four claims, each in one subsection: the bounds are tight; most of the positional error of existing maps is not forced by the tree; at the same topological guarantee, AnchorMTM draws features at their locations; and a budget buys position where conflicts are shallow.

6.1 Data and Measures

Data. Table 2 lists two groups of sequences, and each group serves its own purpose. The *bound suite* consists of seven sequences whose trees, matches and references were saved by one earlier pipeline. Because these trees are fixed, we use them to compute the bounds, to enumerate all leaf orders of small frames, and to measure every compared map frame by frame. The *layout suite* recomputes the trees from the fields and compares complete maps. It reconstructs the four planar inputs of the ST-MTM benchmark [10], so that AnchorMTM is compared on the data of the closest prior work, and adds an hourly $PM_{2.5}$ sequence over northern China [29] and the two TMTM volumes [31, 58]. Ring, Storms, Wildfires and the volumes occur in both groups with different preprocessing, and the daily and hourly $PM_{2.5}$ sequences are different data. Author-processed arrays of the ST-MTM benchmark are unavailable, so the layout suite is a disclosed reconstruction rather than an identical-input reproduction; the supplement gives the preprocessing of every sequence.

Measures. NMLE is the merge error d_{top} of the rendered profile divided by the scalar range, and NROE is the order error of Eq. 2. NPE is the worst displacement of a frame divided by the extent L and averaged over frames; it is the error that a reader of the axis experiences. To credit the goal of ST-MTM, we also report the scale-normalized stress (SNS) [10, 50] and the trustworthiness (TW) [56] of anchor distances against support-centroid distances. The baselines are our implementations of the published methods on the same trees, identities and references. For the four benchmark inputs, the SNS and TW of TMTM and ST-MTM are quoted from the ST-MTM paper (grey in Table 1) and are not ranked against local values. Neither TMTM nor ST-MTM fixes the direction of its axis, so each baseline is reflected once per sequence to agree with the reference.

Modes. Exact is Exact Mode. Balanced is Budgeted Mode with a budget chosen once from 1–10% on the first chronological quarter of each sequence and then frozen: 1, 1, 5 and 10% on Gaussian, Ring, Wildfires and Storms. Hourly $PM_{2.5}$ uses a fixed budget of 5%. The volumes in Table 3 use stored configurations and a $+x$ reference; Figs. 6 and 7 use the default configuration of the bound suite. The supplement lists all settings and the budget selection.

6.2 The Bounds Are Tight

Controlled trees. We first check the bounds where the optimum is known. Two sibling pairs $\{A,B\}$ and $\{C,D\}$ merge at 0.4 below a root at 1, with leaf values 0 and references $(0.2, 0.5, 0.5-d, 0.8-d)$ on a unit canvas; for $0 < d < 0.3$ this is the order A,C,B,D of Fig. 2. Across 43 overlaps d from -0.12 to 0.30 and four widths, we compare the bounds with all 24 leaf orders, each placed by an independent minimax linear program. Topology alone forces exactly $v^* = \max(0,d)/2$. With widths 0.06, gap 0.01 and $\rho = 0.5$, the width bound, our search and the exhaustive optimum coincide at $\tau = \max(0,(d+0.04)/2)$ (Fig. 4a): 12% for points and 14% for intervals at $d = 0.24$. A merge budget removes the conflict exactly where Theorem 3 admits interleaving, at $2\varepsilon = h$ for all four parent gaps h (Fig. 4b).

Real trees. The same agreement holds on real data. In Exact Mode, the interval cost τ of the selected order equals $\mathscr{B}_w$ on 606 of the 614 planar frames of the bound suite, so on these frames no exact-topology order has a smaller worst displacement; at $\varepsilon =$ 10%, the bounded-topology bound is met on 596. On Storms, OLR, daily $PM_{2.5}$ and Ring, every frame has at most eight leaves, and enumerating all leaf orders confirms that AnchorMTM reaches the least order error at every tested budget, except one Storms frame at $\varepsilon = 31.6\%$ (1.9 points). Volumes are harder: the width bound is met on 81 of 102 Cylinder and 27 of 201 Tangaroa frames, and the remaining gap may lie in the bound or in the heuristic search.

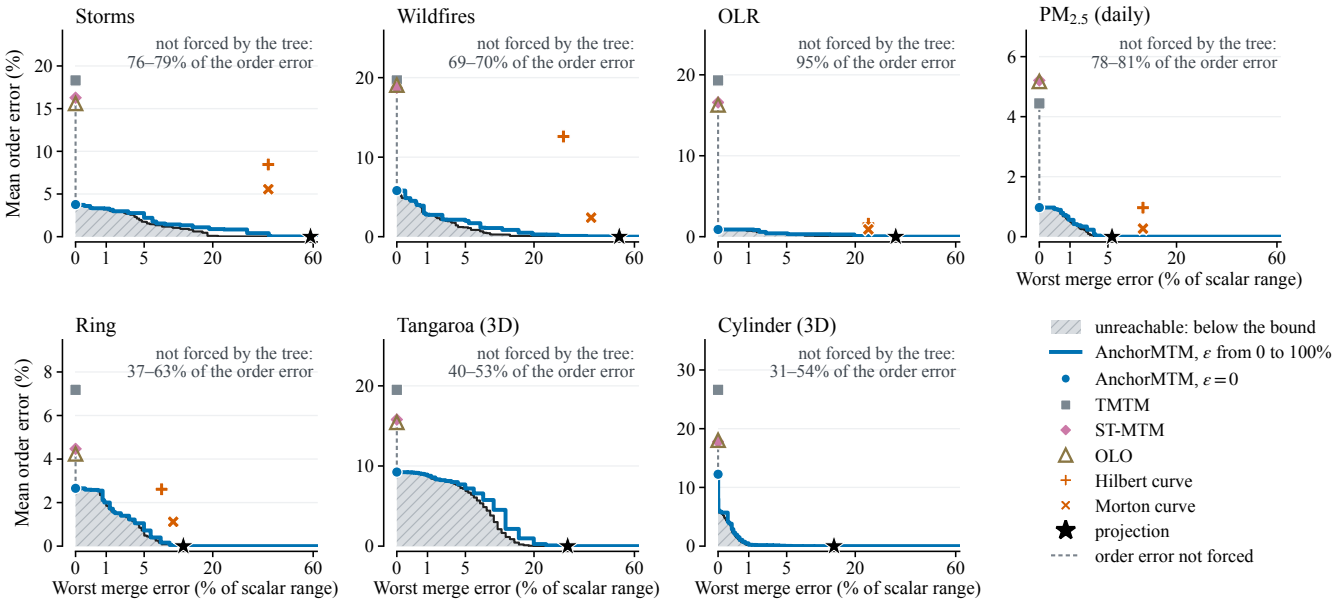


Figure 5: Position–topology plane of the seven bound sequences. Each map is one point, placed by its worst merge error over frames (square-root scale) and its mean order error. No map can enter the hatched region below the bound (Theorem 3 and the flattened bound; Theorem 1 at zero merge error). TMTM, ST-MTM and OLO keep the tree and lie at zero merge error far above the bound; the dashed segment is their order error that the tree does not force, and the label gives its share. Space-filling curves and the projection follow space and break the tree. AnchorMTM (blue line, ε from 0 to 100%; dot, $\varepsilon = 0$) runs along the edge of the region. TMTM, ST-MTM and OLO are shown in the mirror that agrees better with the reference.

6.3 Existing Maps Pay More Than the Tree Demands

Theorem 1 needs no layout, so it measures every map that keeps the tree. On the seven bound sequences, 53–95% of the order error of TMTM and 31–95% of that of ST-MTM lie above ν^*; another order of the same tree would remove this part. Optimal leaf ordering (OLO), which only keeps similar leaves adjacent within the tree, behaves alike (32–95%). The excess therefore comes from choosing the order without a spatial reference, not from one particular method. Figure 5 places every map in the plane spanned by the two errors. Maps that keep the tree sit at zero merge error far above the bound. The projection reaches zero order error only with merge errors of 5.6–59%, and the space-filling curves also break the tree (8–40%). AnchorMTM runs along the edge of the unreachable region, and this edge shows what a budget releases. Where conflicting subtrees merge just below their parent, a budget of 1% halves the forced order error on Wildfires (5.80% to 2.77%) and almost removes it on Cylinder (12.24% to 0.23%); the deep conflicts of Tangaroa keep it above 7% up to $\varepsilon = 5\%$. The forced error also depends on the direction, and it can be computed before a map is drawn: over 400 directions, it spans 4.70–10.14% on Tangaroa and 5.49–28.15% on Cylinder.

6.4 Features at Their Locations

Planar inputs. With the tree kept exactly, Exact lowers NPE relative to TMTM and ST-MTM by 73–96% on all five planar inputs of Table 1 (73.5 and 75.1% on $PM_{2.5}$), and it is also below the Hilbert and Morton curves, which break the tree. Balanced, with a budget of at most 10%, lowers NPE relative to Exact by a further 16.2% on Ring, 11.6% on Wildfires and 64.9% on Storms; on Gaussian it leaves the map unchanged, and on $PM_{2.5}$ the fixed 5% budget lowers NPE by 66.3%. These results show alignment to a chosen reference, not a general gain in quality. ST-MTM reports a higher TW on all four benchmark inputs. On $PM_{2.5}$, where every score is local, ST-MTM also keeps a lower SNS and a higher TW than Exact (0.1897 and 0.8224 against 0.1964 and 0.7884). Against the curves, Balanced improves SNS and TW on Gaussian, Wildfires and Storms; Hilbert is better on Ring (SNS 0.1252 and TW 0.9048 against 0.1334 and 0.8571, with TW from one eligible frame), and both curves keep a higher TW than the 5% map on $PM_{2.5}$.

Table 3: Volumetric comparison with a fixed x reference. Bold/underline mark best/second distinct values per dataset; ties share emphasis.

Dataset	TMTM			AnchorMTM (ours)			
	SNS↓	TW↑	NPE↓	Mode	SNS↓	TW↑	NPE↓
Cylinder	0.2956	0.8017	0.4769	Exact	0.1226	0.8010	0.1773
				Balanced	**0.0329**	**0.9117**	**0.0458**
Tangaroa	0.5255	0.6701	0.3866	Exact	0.1204	0.8213	0.1285
				Balanced	**0.1124**	**0.8408**	**0.0653**

All values are local measurements, including adapted TMTM. TW uses $k = 1$, with coverage 102/102 and 201/201. Mode applies only to AnchorMTM. Configurations are frozen exploratory results, not a blind test; settings appear in the supplement.

One storm. Figure 1 shows what these averages mean for a single feature. As the storm moves from the Atlantic to Scandinavia, its reference rises from 0.08 to 0.86 of the axis. TMTM aligns subtrees by temporal overlap and ST-MTM places them by relative distance, so both keep the storm near the middle, and its band conveys neither the location nor the direction of motion. With the exact tree, AnchorMTM draws the storm on average 1.5% of the axis from its reference, and a 10% budget lowers this to 1.0%. These maps use an hourly archive of the Storms fields; the baselines run on the same 744 fields and trees.

Reading motion. A spatial axis should also show whether two features approach or separate. On the four planar bound sequences with such events (Storms, Wildfires, OLR and Ring; 56–124 events in which the distance between two tracked features changes by more than 2% of the domain diagonal within two frames), An-

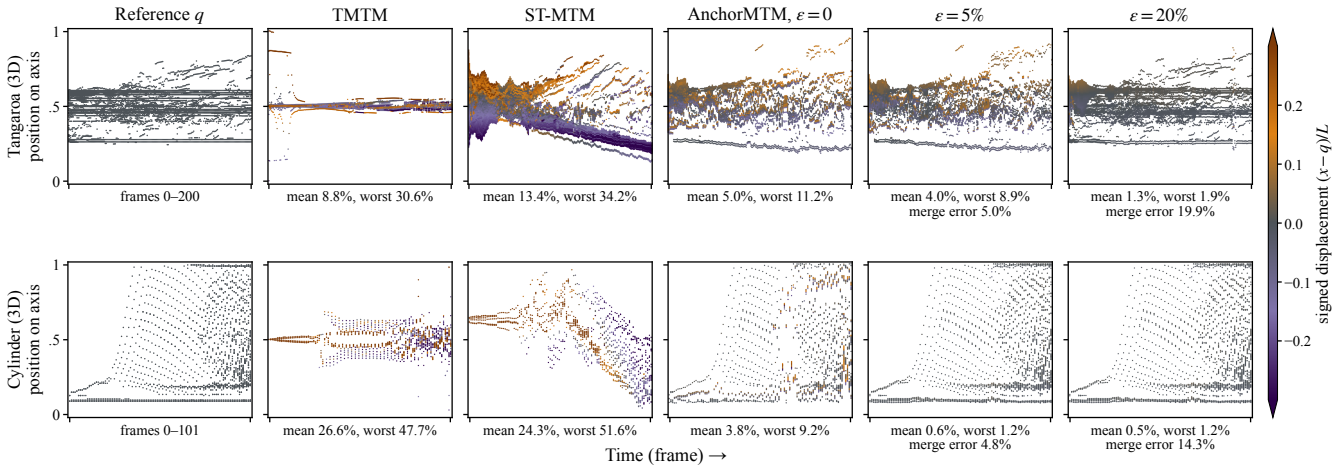


Figure 6: Where each method draws the features of the two volumes. Each dot is one feature in one frame at its position on the axis (fraction of the canvas), colored by its signed displacement from its reference; grey dots lie on their references. The first column shows the references, that is, what a map read as location should show. All methods use the same trees and features, and TMTM and ST-MTM are shown in the mirror that agrees better with the reference. AnchorMTM uses the default configuration with the matched-motion direction (Tangaroa) or the streamwise x axis (Cylinder). Below each panel: mean and mean worst displacement over frames and, for a positive budget, the attained merge error.

chorMTM in Exact Mode shows 5–35% of the events in the wrong direction or not at all, TMTM 39–65% and ST-MTM 21–41%, and neither baseline misreads fewer events on any sequence. Because the motion term of our placement rewards this behavior, the result shows that the axis supports reading motion rather than providing independent evidence.

Volumes. With the stored configurations and the $+x$ reference of Table 3, Exact lowers NPE at zero merge error from 0.477 to 0.177 on Cylinder and from 0.387 to 0.129 on Tangaroa, and Balanced improves all three measures over TMTM; Tangaroa's 33.33% budget and geometry-dominated weights limit attribution. Figure 6 shows the positions behind these numbers under the default configuration. TMTM and ST-MTM gather the Cylinder vortices near the middle of the axis (mean worst displacement 47.7% and 51.6%), whereas Exact already reproduces the shedding pattern of the references (9.2%), and a 5% budget removes almost all of the rest (1.2%). The conflicts in Tangaroa lie deep in the tree: Exact lowers 30.6% (TMTM) and 34.2% (ST-MTM) to 11.2%, and the pattern of the references appears only near $\varepsilon = 20\%$ (1.9%).

Budgets are respected. Exact has zero measured merge error on every input, and Balanced attains at most 0, 0.982, 4.421 and 9.985%, each within its budget. All runs pass the continuous and raster checks without collapsed anchors; Ring needs checked joint integer rounding in 26 Exact and 21 Balanced frames. In the volumetric sweeps, all 1,515 rendered frames were rechecked against the merge matrices of their leaves without a violation. The checks cover the extrema, not every internal label of the augmented tree.

6.5 Where a Budget Acts

A budget limits merge error, not displacement, so its effect on a map depends on how many frames contain a conflict that it can release (Fig. 7). On Storms, 35% of the frames use the allowance at $\varepsilon = 10\%$, in a few runs of consecutive frames. The budget therefore shortens the upper tail: the median per-frame displacement moves only from 1.10% to 0.84%, while the 90th percentile falls from 10.7% to 2.5% and the worst frame from 44.8% to 11.5%. On hourly $PM_{2.5}$, 97% of the frames use the 5% budget and the whole distribution moves: the median falls from 14.5% to 2.1%. The volumes behave alike. At $\varepsilon = 2\%$, 200 of 201 Tangaroa and 98 of 102 Cylinder frames use the budget; the Cylinder median falls from 6.8% to 1.0% already at 2%, and the Tangaroa median from 11.3% to 1.2% at 20%. On Wildfires, the same 7 of 32 frames use the allowance at every budget, and on Ring the median falls from 3.9% to 1.0% while the last frame stays at 10.0% until the budget reaches 63%. On Gaussian, the budget changes no frame. The supplement shows these distributions for all sequences, together with an hourly $PM_{2.5}$ pollution episode read from the map.

7 Discussion and Limitations

Using the bounds. The bounds split the positional error of any merge tree map into a part that the tree forces and a part that its layout chooses. Order error above v^* can always be removed without changing the tree, because midpoint ordering attains v^*. The width-aware and bounded-topology bounds need not be attained, so an excess above them may lie partly in the bound. Because the bounds need only the trees, the references and the widths, an analyst can compute them before drawing a map, to compare candidate directions or to decide whether a budget is worth spending, and any new layout of this kind can be measured against them.

Position and neighborhoods. A fixed spatial reference and a geometry-oriented embedding answer different questions. AnchorMTM supports reading positions and motion along a chosen direction, whereas the higher TW of ST-MTM shows the value of preserving local neighborhoods; neither property guarantees the other. Our results support this task-specific claim, not a general superiority in spatial or temporal fidelity. One axis also carries only one coordinate. Reading two-dimensional positions would require a second map along another direction, with its own bound.

Limitations. Author-processed fields and their small-sample TW protocol are not fully available, so the published SNS and TW cannot support an identical-input significance test, and the two suites differ in preprocessing. The default reference and the tracks are computed offline from the whole sequence, so online use needs a direction chosen in advance, and support overlap is a proxy for correspondence, not a physical identity. The order search is heuristic, and a continuous certificate does not remove raster rounding; we therefore check the rendered merge matrices, but not every augmented internal label. We have not measured perceptual readabil-

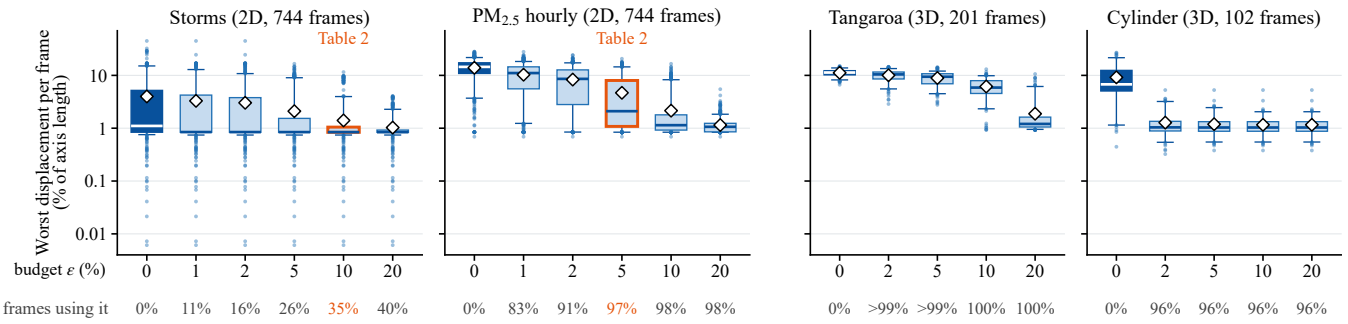


Figure 7: Per-frame worst displacement of AnchorMTM over the merge budget ε on complete sequences (log scale): two planar inputs with the configuration of Table 1 and the two volumes with the default configuration. Boxes: quartiles; whiskers: 5th–95th percentiles; dots: all frames outside the whiskers; diamonds: means, the NPE of the tables; dark: Exact; orange: the budget of Table 1. The bottom row gives the share of frames whose attained merge error is positive; no frame exceeds its budget.

ity or the accuracy of human motion judgments; these require user studies.

8 Conclusion

Giving a merge tree map a spatial axis has an unavoidable positional price, and this price can be computed from the data before a map is drawn. AnchorMTM pays little more than this price. With the tree kept exactly, its leaf order attains the bounds on 606 of 614 planar frames, and it lowers the mean worst displacement of TMTM and ST-MTM by 73–96% on five planar inputs; most of their order error is not forced by the tree. A single merge-error budget, checked on every rendered frame, buys further position where conflicts are shallow, and the bounds tell in advance where it will help.

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